

Switched Network

Switched Networks

- ▶ A **switched network** is a network in which a device called a **switch** connects two or more communication links together.
- ▶ The switch receives data from one link and forwards it to another link whenever needed.

Definition

- ▶ *A switched network is a communication network in which switches are used to forward data from one link to another.*

Role of a Switch

A switch performs the task of:

- ▶ Receiving data
- ▶ Determining the destination
- ▶ Forwarding the data to the correct link
- ▶ In modern communication systems, switching helps multiple users communicate efficiently.

Types of Switched Networks

There are two main types of switched networks:

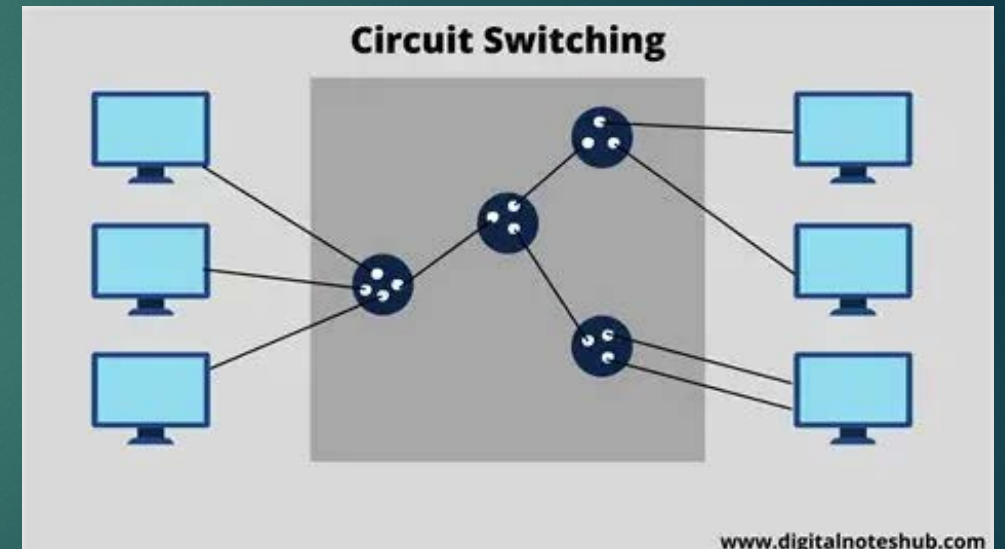
- ▶ **Circuit-Switched Network**
- ▶ **Packet-Switched Network**

Circuit-Switched Network

- ▶ In a **circuit-switched network**, a dedicated communication path called a **circuit** is established between two end systems before communication begins.
- ▶ The path remains reserved during the entire communication session.

Working Principle

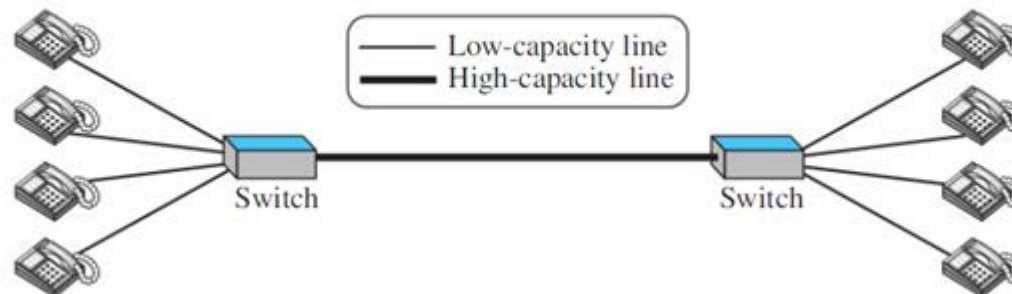
- ▶ A fixed path is created between sender and receiver
- ▶ Resources are reserved for the whole communication period
- ▶ The connection can either be:
 - ▶ Active
 - ▶ Inactive
- ▶ The switch only activates or deactivates the circuit.



Example

- ▶ Traditional telephone networks are classic examples of circuit switching.
- ▶ Telephone ---- Switch ===== Switch ---- Telephone
- ▶ Once a call is established:
- ▶ The communication path remains dedicated
- ▶ Other users cannot use that reserved capacity

Figure 1.6 *A circuit-switched network*



Characteristics of Circuit Switching

Dedicated Path

- ▶ A fixed communication path exists between two devices.

Continuous Communication

- ▶ Data flows continuously throughout the session.

No Storage in Switches

- ▶ Switches only forward signals and do not store data.

Resource Reservation

- ▶ Bandwidth is reserved even if no data is being transmitted.

Advantages of Circuit Switching

- ▶ Continuous transmission
- ▶ Predictable performance
- ▶ No packet delay during communication
- ▶ Suitable for voice communication

Disadvantages of Circuit Switching

Inefficient Resource Usage

- ▶ Reserved bandwidth may remain unused.

Wastage of Capacity

- ▶ Resources are occupied even during idle periods.

Less Efficient for Data Traffic

- ▶ Computer communication is bursty rather than continuous.

Example - Disadvantage

Suppose:

- ▶ Four telephone pairs can communicate simultaneously
- ▶ The main communication line has capacity for four calls

If:

- ▶ Only one call is active

Then:

- ▶ Only one-fourth of the capacity is used
- ▶ Remaining capacity stays unused
- ▶ This makes circuit switching inefficient during low traffic.

Packet-Switched Network

- ▶ In a **packet-switched network**, data is divided into small units called **packets**.
- ▶ Each packet is transmitted independently through the network.

Working Principle

Instead of continuous communication:

- ▶ Data is broken into packets
- ▶ Packets are stored temporarily
- ▶ Then forwarded toward the destination

This method is called:

- ▶ ***Store-and-Forward Transmission***

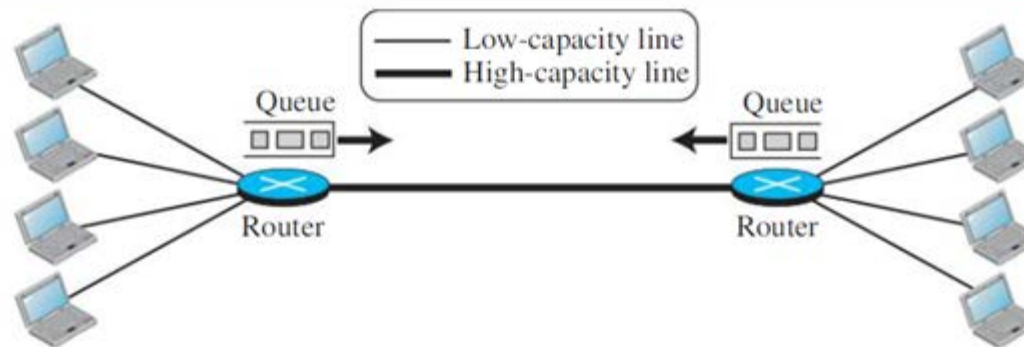
Example

- ▶ Modern computer networks and the Internet use packet switching.
- ▶ Computer ---- Router ===== Router ---- Computer

Routers:

- ▶ Store packets in queues
- ▶ Forward them when the communication line becomes available

Figure 1.7 *A packet-switched network*



Characteristics of Packet Switching

Data Divided into Packets

- ▶ Large messages are split into smaller packets.

Store-and-Forward Mechanism

- ▶ Packets can be stored temporarily before forwarding.

Shared Network Resources

- ▶ Bandwidth is shared among multiple users.

Dynamic Communication

- ▶ Packets may take different paths to reach the destination.

Advantages of Packet Switching

- ▶ **Efficient Utilization**
- ▶ Bandwidth is used only when data is transmitted.
- ▶ **Better Resource Sharing**
- ▶ Multiple users can share the same communication line.
- ▶ **Suitable for Computer Networks**
- ▶ Efficient for bursty data transmission.

Disadvantages of Packet Switching

- ▶ **Delay**
- ▶ Packets may wait in queues during congestion.
- ▶ **Variable Transmission Time**
- ▶ Packets may arrive at different times.
- ▶ **Complex Processing**
- ▶ Routers need memory and processing capability.

Example

Suppose:

- ▶ The main communication line has only twice the capacity of a data line
- ▶ More packets arrive than the line can handle

Then:

- ▶ Routers store packets in queues
- ▶ Packets are forwarded one by one
- ▶ This improves efficiency but introduces delay.

Key Difference

- ▶ *Circuit switching reserves a dedicated communication path, while packet switching divides data into packets and shares network resources dynamically.*

Summary

- ▶ *A switched network uses switches to forward data between communication links. In a circuit-switched network, a dedicated path is reserved for communication, whereas in a packet-switched network, data is divided into packets and transmitted efficiently through shared network resources. Modern computer networks primarily use packet switching because of its better efficiency and flexibility.*

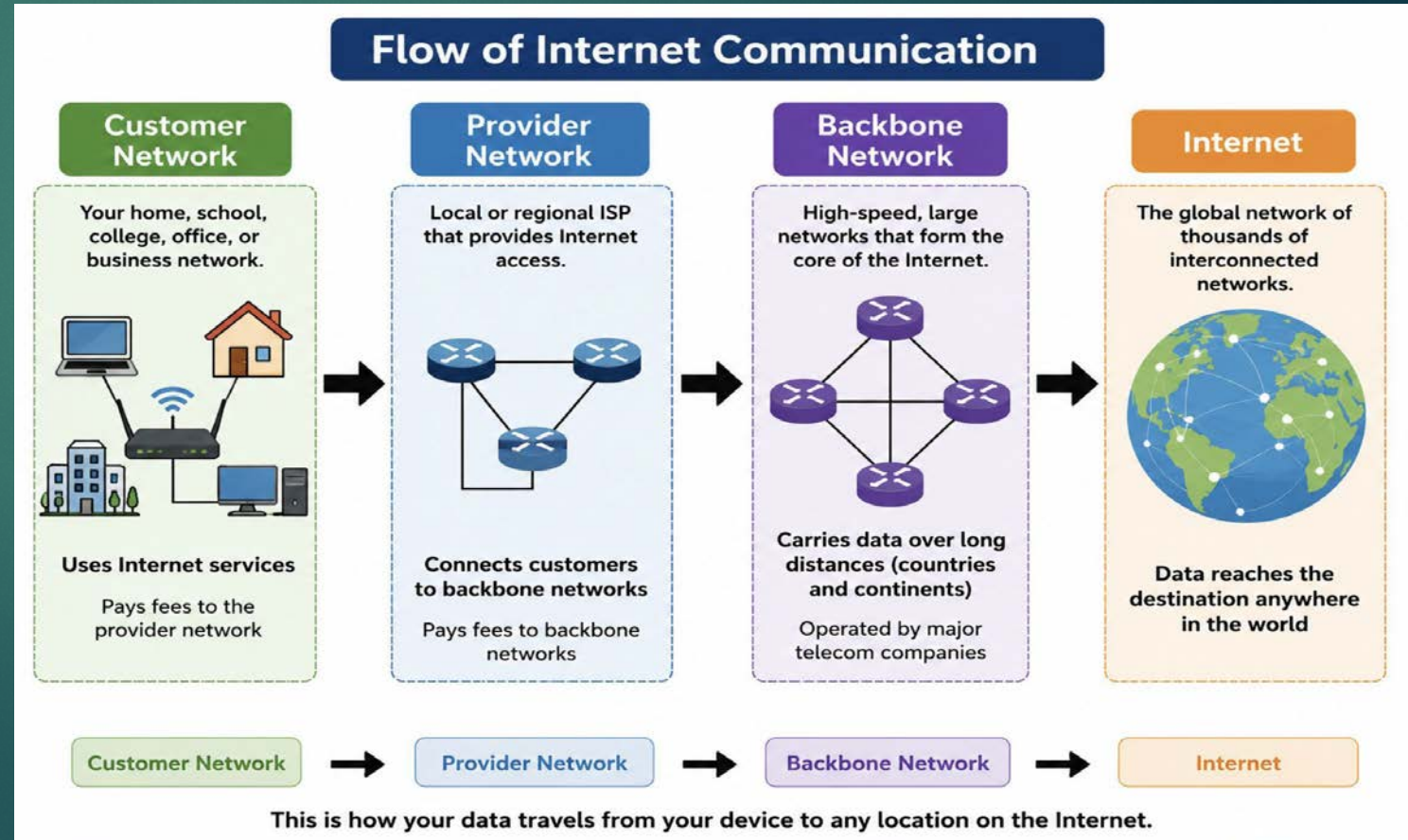
Comparison Between Circuit Switching and Packet Switching

Feature	Circuit Switching	Packet Switching
Communication Type	Continuous	Packet-based
Path	Dedicated path	Shared path
Resource Usage	Reserved for entire session	Used only when transmitting
Efficiency	Less efficient	More efficient
Delay	Minimal after setup	Possible queuing delay
Storage in Switches	No	Yes
Best Suited For	Voice calls	Computer networks and Internet

Structure of the Internet

The Internet consists of three main parts:

- ▶ **Backbone Networks**
- ▶ **Provider Networks**
- ▶ **Customer Networks**



Backbone Networks

- These are the **largest and fastest networks**.
- Owned by major telecommunication companies such as: AT&T, Verizon, Sprint, NTT
- They form the **core of the Internet**.
- Backbone networks are connected through **peering points**.
 - These allow backbone networks to exchange data with one another.

Provider Networks

- Smaller networks that connect to backbone networks.
- They pay fees to use the backbone services.
- They provide Internet access to customers.
- Also called **National or Regional ISPs**.

Customer Networks

- ▶ Customer networks are networks at the edge of the Internet.
- ▶ Examples:
 - ▶ Home networks
 - ▶ School networks
 - ▶ Company networks
- ▶ These networks use the services provided by ISPs.

Internet Service Providers (ISPs)

Both backbone networks and provider networks are called:
Internet Service Providers (ISPs)

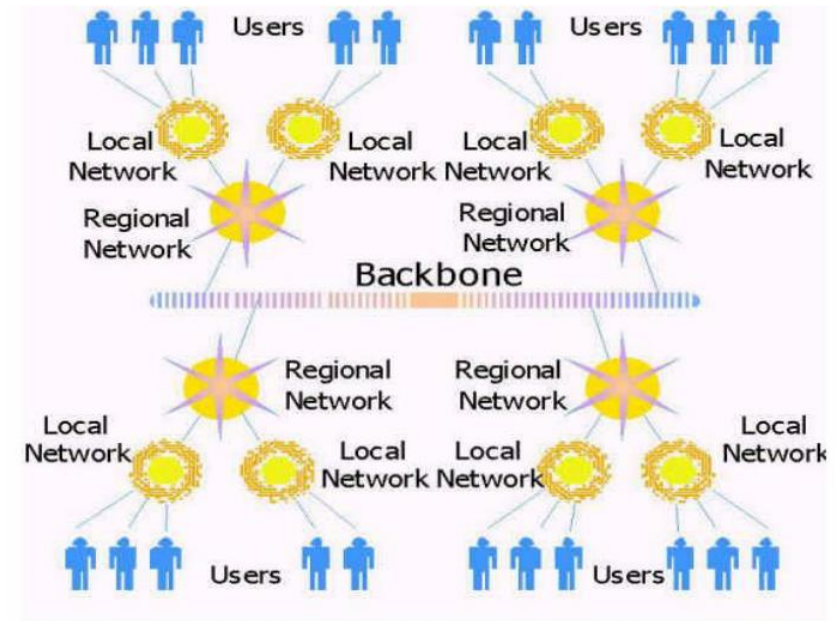
Types of ISPs

Type	Description
International ISP	Large backbone providers
National/Regional ISP	Smaller providers serving customers

Conceptual Structure of the Internet

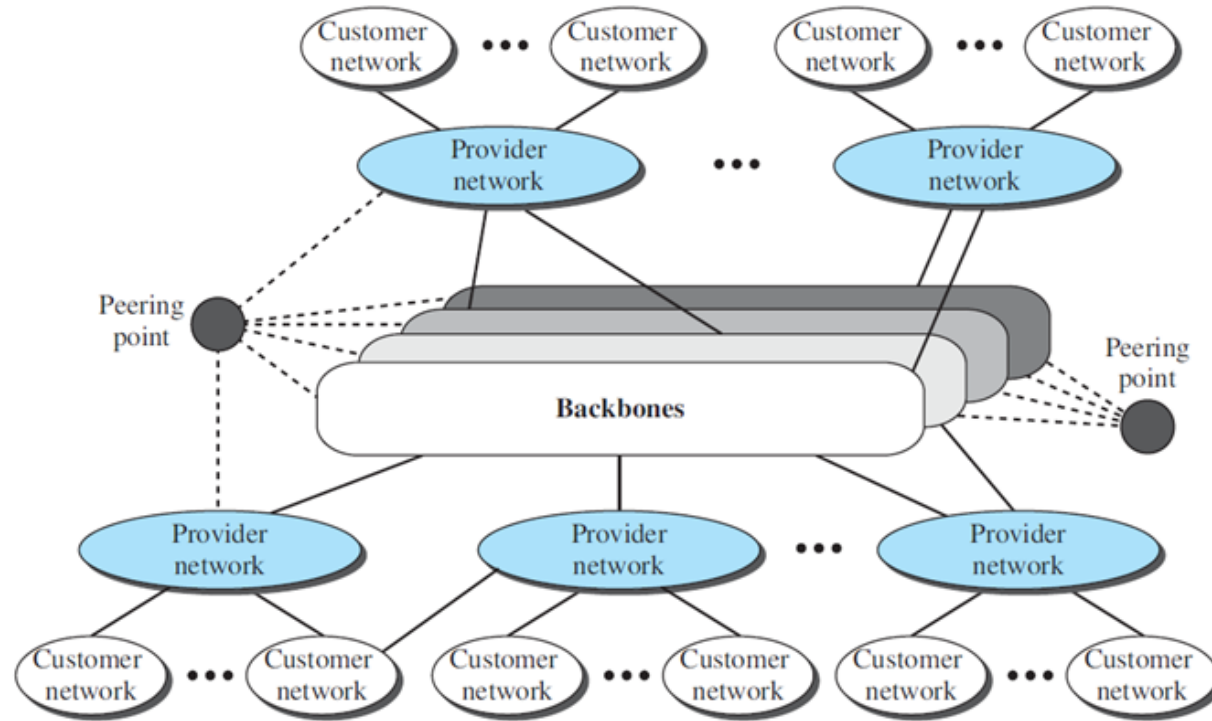
- ▶ Customer Network
|
Provider Network
|
Backbone
|
Provider Network
|
Customer Network
- ▶ The Internet is therefore a huge collection of interconnected networks.

Overall Structure of Internet



Conceptual Structure of the Internet

Figure 1.8 *The Internet today*



Accessing the Internet

- ▶ To become part of the Internet, a user must connect physically to an ISP.
- ▶ This connection is usually established through a:
- ▶ ***Point-to-Point WAN***

Methods of Internet Access

- ▶ **Using Telephone Networks**

- ▶ Most homes already have telephone connections, so Internet access can be provided through telephone lines.

- a) Dial-Up Service**

- ▶ Uses a modem
- ▶ Converts data into voice signals
- ▶ Computer dials the ISP like a telephone call
- ▶ **Limitations**
 - ▶ Very slow
 - ▶ Telephone line cannot be used for voice calls simultaneously

Methods of Internet Access

b) DSL (Digital Subscriber Line)

- ▶ DSL provides:
- ▶ Higher Internet speed
- ▶ Simultaneous voice and data communication
- ▶ It is faster and more efficient than dial-up service.

Using Cable Networks

- ▶ Cable TV networks can also provide Internet access.
- ▶ **Features**
- ▶ Higher speed connection
- ▶ Shared bandwidth among neighbours
- ▶ The speed may vary depending on network usage.

Using Wireless Networks

- ▶ Wireless technology allows Internet access through:
- ▶ Wi-Fi
- ▶ Wireless WANs
- ▶ Mobile networks
- ▶ Wireless access has become very popular due to mobility and convenience.

Direct Internet Connection

- ▶ Large organizations and universities may directly connect to the Internet by:
- ▶ Leasing a high-speed WAN
- ▶ Connecting to a regional ISP

Example:

- ▶ Large universities with multiple campuses

Summary

- ▶ *The Internet is a worldwide internetwork made up of interconnected backbone networks, provider networks, and customer networks. Users access the Internet through ISPs using telephone lines, cable networks, wireless networks, or direct high-speed connections. The Internet enables global communication and resource sharing on a massive scale.*